



AGAMI Aviation
Products and Services



Agami delivers advanced, integrated airside infrastructure solutions tailored to the evolving needs of the Aviation, Military, and Aircraft Maintenance & Repair Organization (MRO) sectors. Specializing in Aircraft Ground Support Utility Systems, we provide centralized and efficient delivery of essential services, including Pre-Conditioned Air (PCA), Ground Power (GPU), potable water, sewage, blue water systems, fume extraction systems, and in-ground solutions for aircraft ground operations.

In addition, Agami collaborates with leading international partners to design and deliver advanced aircraft maintenance infrastructure, including aircraft docking systems, engine test stands, and a wide range of customized MRO solutions. Engineered for precision, safety, and efficiency, our docking systems provide optimal access for maintenance operations across various aircraft types, while our engine test stands ensure reliable and controlled performance testing under demanding conditions.

Through its global partnerships, Agami also offers specialized jet blast protection solutions. Our jet blast deflectors and blast fences are designed to protect critical airport operational areas from high-velocity jet exhaust, enhancing safety across aprons, taxiways, and runways. We provide a comprehensive range of solutions, including fixed and movable jet blast deflectors, as well as noise and blast-deflecting screens.

Our solutions are engineered to enhance operational efficiency, improve safety, and support environmentally responsible operations. With a comprehensive portfolio spanning in-ground pit systems, ground-mounted units, Passenger Boarding Bridge (PBB)-integrated solutions, aircraft docking systems, engine test stands, and specialized jet blast protection solutions, we enable seamless aircraft servicing and maintenance across contact gates, remote stands, and MRO facilities.

Driven by innovation, reliability, and sustainability, Agami supports next-generation airports by delivering smart infrastructure that reduces apron congestion, minimizes emissions, and accelerates aircraft turnaround, thus creating safer, cleaner, cost effective, and more efficient airside environments.



OUR PRODUCTS



PRE-CONDITIONED AIR UNIT



400 Hz GPU



28 V DC GPU



POP UP PIT SYSTEM



HATCH PIT SYSTEM



LOAD BANK



MOBILE GPU



AERIAL JIB PLATFORMS



JET BLAST DEFLECTOR



INTEGRATED AIRCRAFT GROUND SUPPORT UTILITY SYSTEM

PCA | GPU | Potable Water | Sewage | Blue Water

Modern airport operations require reliable, efficient, and environmentally responsible aircraft servicing infrastructure. An Integrated Aircraft Ground Support Utility System provides centralized, safe, and efficient delivery of essential aircraft services including Pre-Conditioned Air (PCA), Ground Power Units (GPU), Potable Water Supply, Sewage Disposal, and Blue Water Services.

These systems can be deployed through in-ground pit systems, ground-mounted units, or Passenger Boarding Bridge (PBB) mounted equipment, depending on the operational requirements of contact gates, remote aprons, and Maintenance, Repair & Overhaul (MRO) facilities.

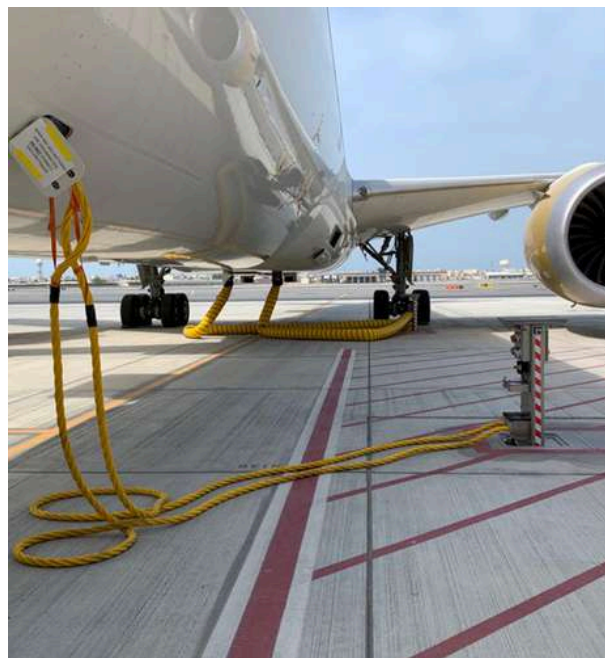
SYSTEM CONCEPT

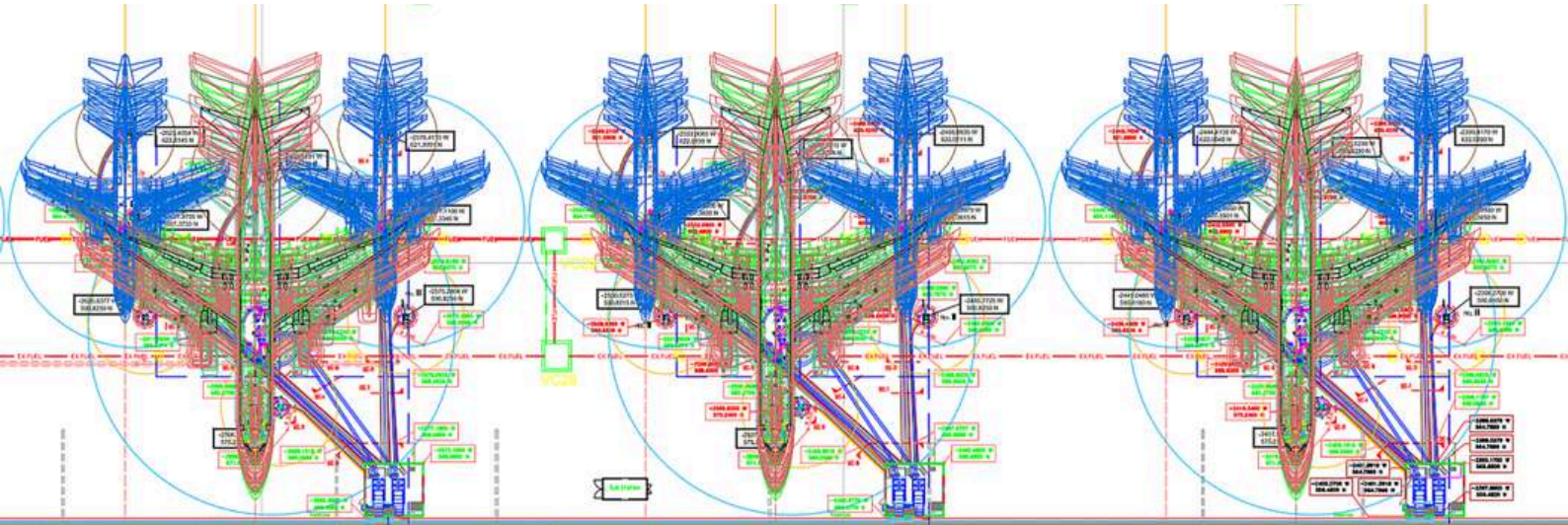
The integrated concept focuses on centralized / Fixed generation, efficient distribution, and safe aircraft interface for all key ground utilities. By minimizing the reliance on mobile diesel equipment and optimizing service points at aircraft stands, the system improves operational efficiency, reduces emissions, and enhances safety on the apron.

THE SOLUTION TYPICALLY INCLUDES:

- Pre-Conditioned Air (PCA) for aircraft cabin cooling / heating while on ground
- Ground Power Units (GPU) supplying electrical power to aircraft systems
- Potable Water Supply for onboard drinking water systems
- Sewage Disposal System for aircraft waste removal
- Blue Water System for lavatory flushing services

These services are delivered via strategically located connection points, allowing rapid turnaround while maintaining a clean and organized apron environment.





SMART INFRASTRUCTURE FOR SMARTER AIRPORTS

Agami specialized in the design, engineering, and integration of airport and MRO systems for civil and military aviation. We deliver end-to-end, sustainable solutions that enhance operational efficiency, safety, and environmental performance across modern airport infrastructure.

With strong domain expertise, Agami supports governments, aviation authorities, airports, and MRO facilities through tailored engineering solutions, ensuring seamless coordination with EPC contractors and stakeholders across the project lifecycle.

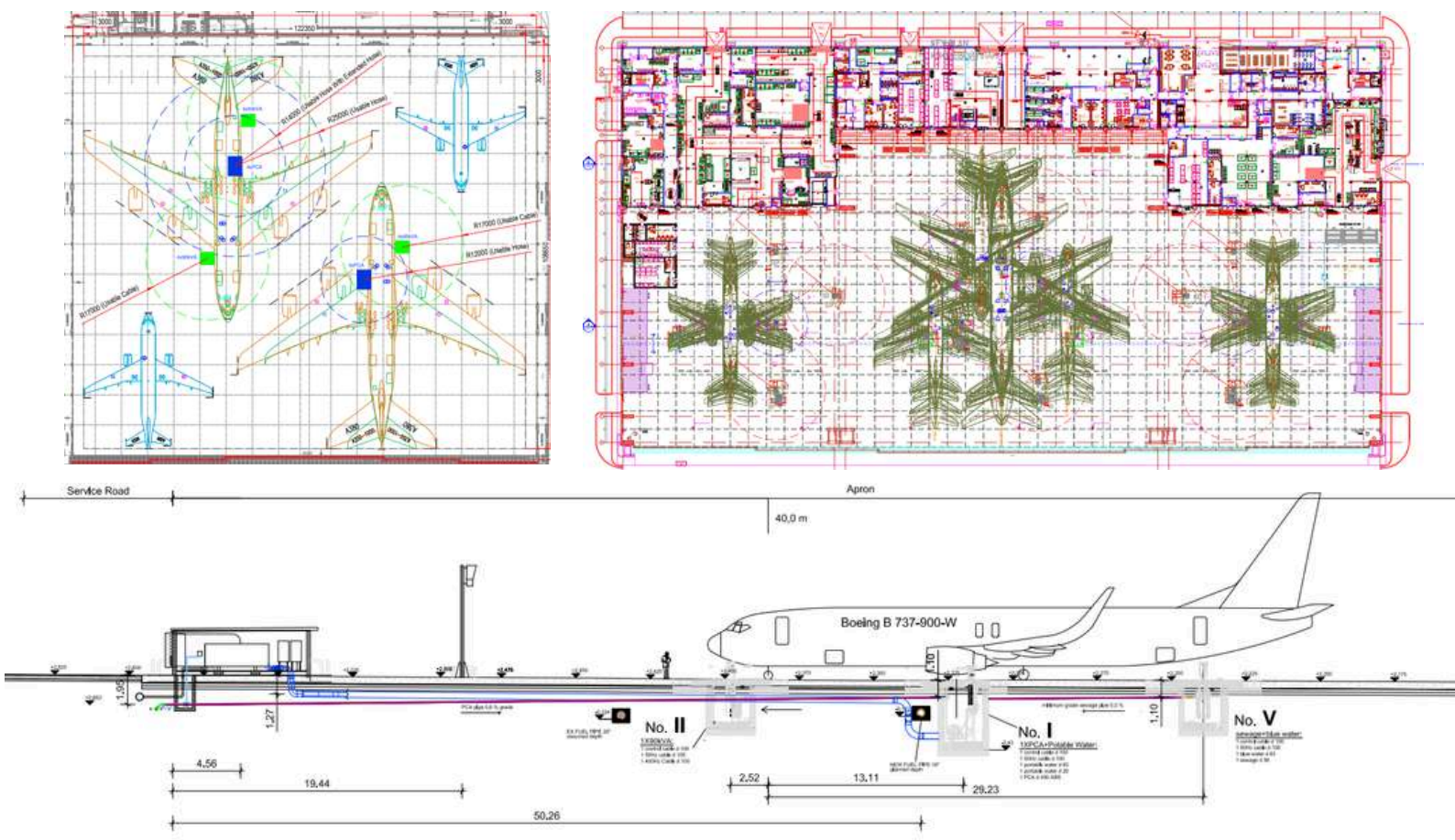
Solutions offered

- 400Hz Ground Power Systems (GPU)
- Pre-Conditioned Air (PCA) Systems
- Potable, Sewage & Blue Water Systems
- In-ground and above-ground service solutions
- Fume extraction systems
- Aircraft docking systems
- Maintenance platforms & access systems
- Engine handling systems

Scope of Services

- System design & engineering for airport and MRO infrastructure
- Integration of airside systems and utilities for aircraft stands
- Project management & technical advisory
- Modernization of existing facilities with advanced technologies
- Airport, Airside & MRO GSE Solutions

Agami solutions are engineered for high reliability, safety compliance, and minimal downtime, delivering efficient performance across both airport operations and MRO environments.





GROUND COOLING TO ENSURE COMFORT OF EVERY PASSENGER...

Agami's partner offers a comprehensive range of Central and Point-of-Use PCA systems, including fixed, mobile, above-ground, in-ground, and subfreezing units, all meeting or exceeding industry performance standards.

Our portfolio includes Central (Hybrid CW+DX) and Point-of-Use (DX) systems, fully compliant with AHM974, AHM997, and ASHRAE standards. Our PCA systems were the first in the world to receive AHM997 approval, enabling aircraft APUs, including Airbus A380, to be safely switched off, even under harsh ambient conditions in Asia and the Middle East.

Our engineers optimize system design, power requirements, airflow distribution, and temperature control from PCA outlets to aircraft inlets, ensuring reliable, efficient, and precise cooling for all aircraft types.

PCA200DX – PRE-CONDITIONED AIR (PCA) UNIT

The point or use DX type PCA200DX is a high-capacity Pre-Conditioned Air (PCA) unit engineered by our partner for demanding airport environments operating under extreme ambient conditions

Executed at Dubai International Airport, this system represents a first-of-its-kind single-unit installation capable of delivering high cooling performance suitable up to Code-F aircraft.

The PCA200DX is a high-capacity Pre-Conditioned Air (PCA) Designed for reliable operation in harsh climatic conditions, the PCA200DX demonstrates our advanced engineering capability and commitment to deliver robust, energy-efficient ground support solutions for high-traffic international airports.



PBB PCA – PRE-CONDITIONED AIR UNIT

The PCA is a high-performance Pre-Conditioned Air unit designed for efficient aircraft cabin cooling under hot and tropical ambient conditions. Deployed at high temperature tropical zone like Delhi International Airport and Dubai Airport, the system is capable of delivering outlet air temperatures as low as -8°C, ensuring rapid cooling performance for Code-C to Code F aircraft.

The PCA is engineered for reliability and operational efficiency, addressing performance limitations commonly encountered in conventional PCA systems and supporting improved passenger comfort and aircraft turnaround time.





ULTRA SUB FREEZING PCA (-25°C)

Our Ultra Sub Freezing PCA Unit is an advanced, centralized chilled-water-based Pre-Conditioned Air (PCA) system engineered to deliver high-performance aircraft cooling across all aircraft categories from Code-C to Code-F.

Designed for maximum operational flexibility, the system provides both sub-freezing and non-subfreezing conditioned air and can simultaneously serve multiple aircraft with independent control of temperature, airflow, and pressure.

This USFPCA can deliver the outlet air temperature of -25°C blast performance at extreme hot ambient condition of upto 50°C.

Compliant with AHM997 and AHM974 standards, the system integrates seamlessly with Gate-PLC and BMS platforms for monitoring, control, and energy management. Its compact, acoustically enclosed dual blower design ensures low noise operation and suitability for modern airport gate configurations, including MARS stands.



MOBILE PRE-CONDITIONED AIR (PCA) UNIT

Our Mobile Pre-Conditioned Air (PCA) Unit is designed to provide fast and effective aircraft cabin cooling during ground operations, even in hot and tropical climates above 45 °C.

The unit delivers outlet air temperatures as low as -8 °C, ensuring quick cabin temperature reduction for both Code C to Code F aircraft. It is ideal for passenger boarding, aircraft turnaround and maintenance operations, while helping reduce APU fuel consumption, emissions and apron noise.

Built on a strong towable trailer platform, the unit is equipped with heavy duty GSE solid cushion tyres for reliable and low-maintenance airport operation. It can be easily maneuvered by an airport tractor and positioned conveniently near aircraft stands, even in limited apron space.





PURE POWER ABSOLUTE SILENCE

Clean, Reliable Aircraft Ground Power

Agami's partner is a leader in the design and manufacturing of advanced power solutions for aviation and military applications. With decades of expertise, we deliver efficient, reliable, and high-performance 400Hz ground power systems tailored to the needs of modern airports and aircraft.

Our portfolio includes centralized and point-of-use systems, such as solid-state frequency converters, rotary motor generators, and mobile diesel-powered GPUs. Supporting capacities from 5 kVA to 180 kVA, our solutions can be deployed as ground-mounted units or Passenger Boarding Bridge (PBB)-mounted systems with integrated cable management.

Built with military-grade components and tested under extreme conditions, our systems ensure durability and consistent performance. They are fully compatible with next-generation aircraft, including Airbus A350, A380, and Boeing 787.

Our systems comply with DFS-400 and MIL-STD-704F standards, exceeding ISO 6858 requirements, ensuring reliable performance at the aircraft connector.

400 Hz GPU Efficient Ground Power Operations

When aircraft are parked with engines off, our Frequency Converters provide a stable 200V/400Hz power supply to support onboard systems. Connected to standard three-phase mains power, the system delivers precise output on demand, optimizing energy efficiency.

In addition to 400Hz systems, we also offer 28V DC GPUs for aircraft applications, known for their simplicity, robustness, and high performance.

Advanced synchronization ensures seamless power transfer without interrupting critical onboard functions such as lighting, air conditioning, and avionics.



Smart Integration & Control

All our GPUs feature built-in Modbus communication, enabling seamless integration with Agami OpsEye, an IoT-based platform for centralized monitoring and control of gate equipment.

- Real-time system visibility
- Faster fault detection and troubleshooting
- Improved operational efficiency
- Sustainable & Cost-Effective

Key Features

- Clean Energy: Unity Power Factor (0.99) and THDi < 5% ensure high-quality, aircraft-safe power
- High Efficiency: > 94% efficiency for reduced energy consumption and superior ROI
- Silent Operation: < 65 dBA @ 1m for a quieter, safer working environment
- Rugged Design: IP54-rated enclosure for reliable performance in all weather conditions



IN-GROUND EFFICIENCY.... ELEVATING ABOVE-GROUND PERFORMANCE.

Agami in-ground pit systems provide a proven, cost-effective solution for clean apron and MRO operations. Designed for both single aircraft stands and Multiple Aircraft Ramp System (MARS) gates, these systems integrate essential services including PCA, GPUs, potable water, and sewage connections within a compact, ground-level solution.

Our in-ground pop-up and hatch pit systems offer user-friendly storage for 400Hz cables, PCA hoses, and additional utilities such as 50/60Hz outlets, ensuring safe and organized ground operations.

Positioned close to aircraft service receptacles, these systems minimize cooling losses, pressure drops, and voltage drops, enabling faster, more efficient aircraft turnaround while significantly reducing APU usage at parking stands.

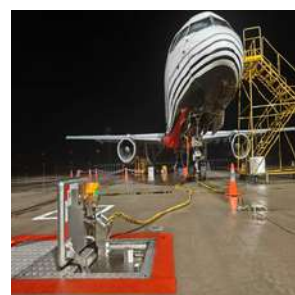
Built with heavy-duty covers and installed at key apron locations, Agami in-ground pits enhance operational efficiency, reduce tarmac congestion, and improve overall safety by minimizing the risk of ground handling incidents.

POP-UP PIT SYSTEM

The Pop-Up Pit System features a precision-engineered, counterweight-assisted lifting mechanism that allows ground personnel to effortlessly raise the service column to an optimal ergonomic working height.

HATCH PIT SYSTEM

The Hatch Pit System is designed for simple, direct access to ground support utilities through a single-touch or assisted opening mechanism.





WET SERVICE SYSTEMS

According to the "Point of Use" philosophy, the following media will be serviced through underground pit systems to the aircrafts:

- Potable Water Supply
- Blue Water (Flushing Water) supply
- Sewage removal from toilets

The ground operations can be improved to achieve faster turnaround times, reduced CO2 emissions, improved safety and reduced accidents on the apron.

- Eliminates the need for mobile equipment to service the aircraft
- Reduces congestion of the apron
- No air and noise pollution
- Reduces operational and maintenance costs
- Reduces security and safety issues
- Easier access to the aircraft (especially in case of emergency) as the left side of the aircraft is free of vehicles

BLUE WATER SYSTEM

The wastewater toilet tanks of the aircraft have to be flushed with a disinfectant solution ('blue water').

The water mixture is produced fully automatically in the plant room and its quality is monitored online in real time. It is then pumped via the underfloor pipeline network in the apron up to the individual pit systems.



POTABLE WATER SYSTEM

The fresh water vessels on board of the aircraft will be filled with potable water as per WHO standard, airline and local regulations.

The basic treatment includes disinfection, storage, pressure booster and circulation systems.

If required raw or brackish water treatment and chilling of potable water can be provided as well. The process is operated and monitored automatically.



VACCUM STATION

The vacuum station will be located in a central plant room, which is usually in the Terminal basement or inside separate technical building.

The wastewater from the aircraft is collected in a wastewater vessel and then conveyed with wastewater discharged pumps into the municipal sewer system or to the sewage treatment plant of the airport.





ADVANCED HANDLING SOLUTIONS FOR AVIATION & SPACE:

At Agami, we collaborate with international partners to transform complex challenges into smart, efficient solutions. Our innovative systems streamline operations, enhance productivity, and ensure seamless material flow across industries such as intralogistics, surface treatment, aviation, and space.

Agami delivers high-performance, tailor-made handling and access solutions for aviation and space applications, including aircraft manufacturing, MRO operations, painting, engine handling, and space program support. Designed for precision, safety, and efficiency, our systems help optimize workflows while ensuring full compliance with industry standards.

AIRCRAFT DOCKING SYSTEMS:

Our docking systems provide safe and efficient access to all aircraft areas, incorporating advanced safety features such as folding railings and fall-prevention systems. Solutions include wing, fuselage, nose, tail, and engine docks, enabling comprehensive maintenance and assembly operations.



TELEPLATFORMS

Teleplatform systems offer exceptional flexibility, allowing access to various aircraft types and positions within hangars. Integrated with overhead crane systems, they enable efficient inspection and maintenance, even for mixed fleets and limited-access configurations.



ENGINE HANDLING SOLUTIONS

Agami provides a wide range of precision-engineered engine handling and transport systems, ensuring safe and efficient maintenance of high-value aircraft engines. Solutions include adjustable workstations, integrated transfer systems, hoist carriers, and monorail systems for streamlined assembly and disassembly processes.





JET BLAST DEFLECTORS & BLAST PROTECTION SOLUTIONS

Jet blast deflectors and blast fences protect airport operational areas from high-speed jet engine exhaust, typically installed between aprons, taxiways, runways, and other critical zones to ensure safe operations while minimizing space requirements. Agami offers a complete range of solutions, including fixed and movable deflectors, blast fences, and noise screens, designed for full-power engine operations.

These solutions can be permanent or portable, tailored to each airport project's requirements. Structures are engineered based on site-specific conditions, with radar-neutral and ILS-compatible options available through fiberglass deflectors.

TYPES OF JET BLAST DEFLECTORS

- Solid Deflectors: Solid, angled panels that redirect exhaust vertically ($\approx 70^\circ$ – 90°), robust enough for full-power engine run-ups.
- Mesh Deflectors: Metallic mesh panels that redirect jet blast while allowing partial visibility; configurable in inclined or upright orientations.
- Transparent Deflectors: PMMA (acrylic) lamellas providing full visibility without compromising structural integrity; interchangeable with mesh panels.
- Fiberglass (Radar-Neutral) Deflectors: Non-metallic structures compatible with radar, ILS, and other navigational systems for sensitive operational areas.





MOBILE GSE

GROUND POWER UNIT

Our diesel engine / electric motor driven Ground power units comply to latest EPA regulations with low noise levels and are compact in size to avoid cluttering around aircrafts.

It comes with a digital control systems - touch screen panel for monitoring of operational parameters with in - built Agami OpsEye - a feature rich fleet management software.

Available in Towable Trailer mounted / Truck mounted / Self propelled configurations.



AIR START UNIT

Our Air start units are capable of starting aircraft engines up to GE 90, the world's most powerful commercial jet engine to date, made with screw compressors that are the most efficient in the market.

We design and manufacture storage type air starter units as well. Auto adjusted air flow depending on aircraft requirement via digital control system and in - built Agami OpsEye.

Available in Towable Trailer mounted / Truck mounted / Self propelled configurations.

AIR CONDITIONING UNIT

Our direct / electric driven Air conditioning unit provide quick cooling or heating to an aircraft's cabin on ground. We manufacture ACUs for all types of aircraft.

Provide fully automated temperature control via digital control system (touch screen) and in built Agami OpsEye.

Option: Shore power transfer switch / 28.5V DC output

Available in Towable Trailer mounted / Truck mounted / Self propelled configurations.



28V STATIC GROUND POWER UNIT

Meets MIL, AHM & ARP standards

Output DC voltage independent of 10 %

Output current limit setting is adjustable over the range of 200 to 2000 amps

Convenience receptacle for lights, electric drill, or test instruments

ENGINEERED TO PURGE.... DESIGNED TO PROTECT....

AIRCRAFT FUEL TANK VENTILATION SYSTEM:

Aircraft fuel tanks must be properly ventilated before any maintenance work to ensure safety. Fuel vapors are highly flammable and toxic, and if they accumulate inside the tank, they can create serious fire, explosion, and health risks.

A dedicated ventilation system safely removes these fumes, creating a safe working environment for maintenance personnel and protecting equipment. It also helps prevent ignition caused by tools, electrical devices, or static discharge.

VENTILATION AND FUME EXTRACTION ARE ESSENTIAL TO:

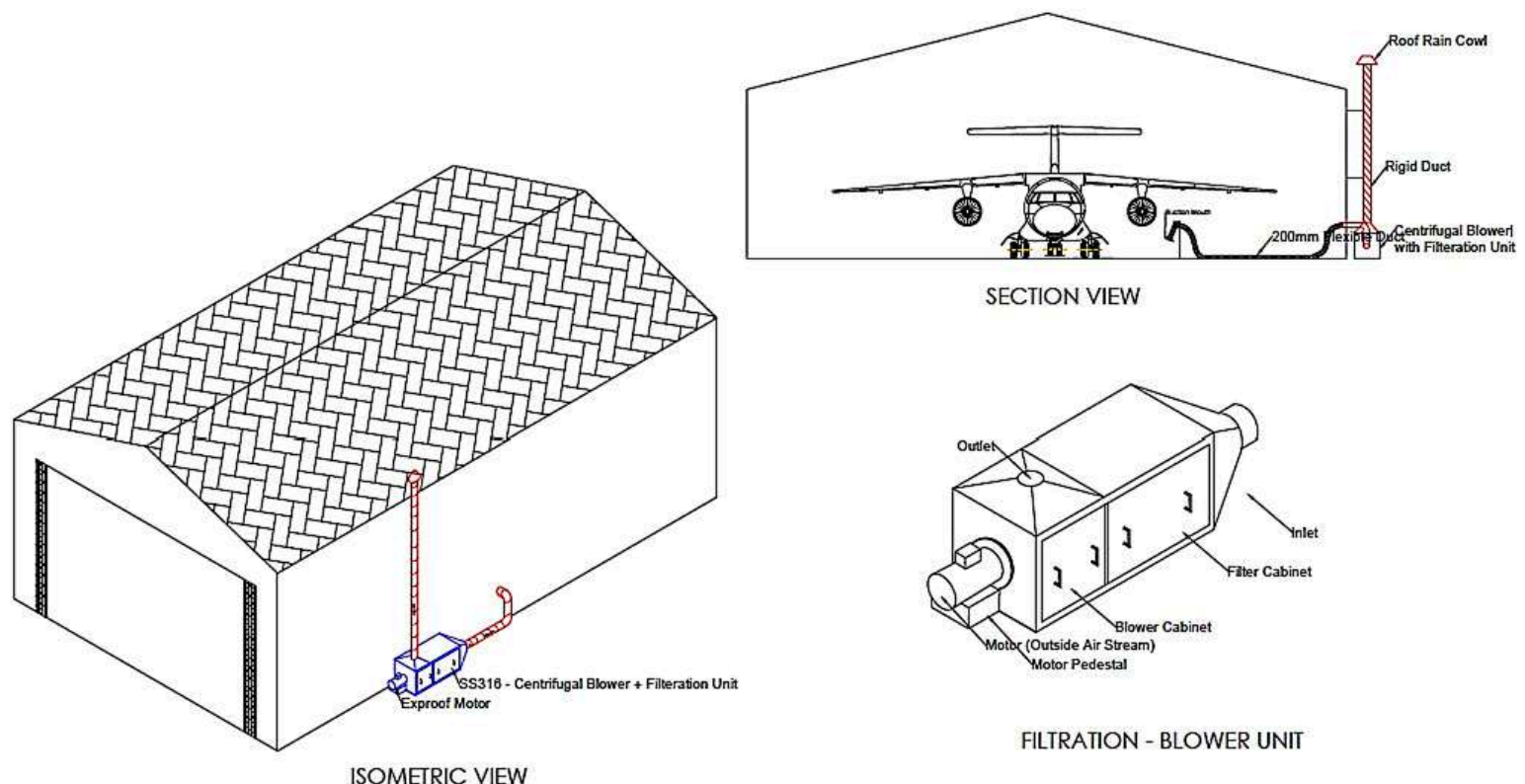
- Prevent fire and explosion risks
- Protect worker health
- Comply with aviation safety standards

Our solution provides fixed independent ventilation units alongside the hangar or through Pits for efficient operation.

The proposed system uses an explosion-proof centrifugal fan designed to safely extract hazardous fumes during aircraft maintenance. The system includes a two-stage filtration unit made from corrosion-resistant SS316.

SAFETY & COMPLIANCE

- Designed for Hazardous Area Zone 1
- Handles Group IIA/IIB gases (IECEx compliant)
- Flameproof (Ex) motor and electrical components
- Anti-static construction with proper grounding



FUMES EXTRACTION SYSTEM TYPICAL DRAWING

SMART MONITORING FOR AIRPORT GROUND SUPPORT EQUIPMENT

Agami OpsEye

Agami OpsEye is a cloud or AODB enabled monitoring platform designed for centralized supervision of airport ground support equipment such as PCA, GPU, PBB, and VDGS systems. The platform collects real-time operational data, equipment status, and fault information from each aircraft stand and displays it through an intuitive graphical dashboard, enabling efficient monitoring, faster troubleshooting, and improved operational reliability across airport operations.

The system also provides advanced capabilities including remote fault diagnosis with 3D equipment visualization, historical performance tracking, maintenance scheduling, spare parts management, and customizable operational reports.

Additional features such as automatic billing for equipment usage, ROI analysis, CCTV monitoring, and integrated communication help airports optimize asset utilization, streamline maintenance, and enhance overall operational efficiency.



KEY CAPABILITIES

- **Real-time monitoring:**

Live tracking of all connected equipment with detailed insights into performance, utilization, and faults.

- **Remote Diagnostics & 3D Visualization:**

Advanced fault detection with component-level identification using interactive 3D models for faster troubleshooting.

- **Maintenance & Asset Management:**

Comprehensive maintenance scheduling, service history tracking, and spare parts inventory management.

- **Automated Billing & ROI Analysis:**

Automatic calculation of equipment usage, airline billing, and return on investment for optimized financial performance.

- **Reporting & Analytics:**

Customizable reports with filtering by date, equipment, or stand, supporting data-driven decision-making.

- **Integrated Communication & CCTV Monitoring:**

Built-in communication tools and live CCTV monitoring for enhanced coordination and security.

- **Operator Console & Access Control:**

The platform supports advanced authentication methods, including password protection, biometric integration, and secure login protocols, safeguarding sensitive operational data.





AGAMI Aviation

PO-Box 293624, Office # 52, White Future,
1st Floor, Madina Mall, Beirut St,
Muhaisnah 4, Dubai, UAE

+971 45 47 3252
info@agamiaviation.com
www.agamiaviation.com
www.agamiengineering.com